

DERIVATIVES 101 • DECK 1

KODA & KODD

Knock-out discount accumulators and decumulators.

The biggest structured product in Hong Kong, and the one that has hurt clients most.

IN ONE SENTENCE

A contract to keep buying, a little every day, at a discount

The client is obliged to buy a fixed quantity of the underlying every trading day, at a price below today's spot, until the price rises through a knock-out barrier.

Two catches. Below the strike, the daily quantity doubles. And the knock-out caps the upside at a few percent, while the downside runs all the way to a share price of zero.

Obligation

Not a right. The client must buy every single day, whatever has happened.

Discount

The strike sits 5–15% below spot. That discount is option premium, not a gift.

Gearing

Below strike the daily size doubles. You buy most where you are most wrong.

Knock-out

A rally ends the trade early, with a small profit. That is the good outcome.

Hong Kong investors renamed it in 2008. 'Accumulator' became 'I kill you later' — a homophone that stuck, and is now used in the regulator's own investor-education material.

Reading an accumulator term sheet

Reference Underlying	A single listed share, an FX pair, or occasionally an index
Reference Spot	USD 320.00 (100%)
Forward Price	USD 288.00 (90%) — the price the client is obliged to buy at
Knock-Out Price	USD 336.00 (105%), on the closing price, from the trade date
Underlyings per Day	10
Leverage	2 — so 20 a day when the close is below the Forward Price
Settlement Frequency	Monthly, in arrears, physical delivery
Number of Observation Periods	12
Guaranteed Observation Period(s)	2 — the first two months accumulate whatever happens
Max. Scheduled Trading Days	250
Max. Total No. of Underlyings	$250 \times 20 = 5,000$
Max. Exposure	$5,000 \times \text{USD } 288.00 = \text{USD } 1,440,000$

Three numbers to check first

Max. Exposure

Not the margin. This is the real size of the obligation — USD 1.44m against roughly USD 432,000 posted.

Forward Price to Knock-Out

288 to 336 is the entire window in which this trade works. Fifteen percent of spot.

Gearing multiple

2x is standard. 3x exists. It changes the maximum loss, not the maximum gain.

THE CORE PICTURE

What one observation day is worth

Client P&L for a single observation day



- The whole trade is this chart, repeated 120 times.
- Above the Forward Price the client makes a little. The most any single day can earn is the gap up to the Knock-Out Price — and reaching it ends the trade.
- Below the Forward Price the slope doubles. There is no barrier and no airbag on this side — the only floor is a share price of zero.
- Note where the chart is truncated on the right and open-ended on the left. That asymmetry is the product.

Ten units bought on any day the close is at or above the Forward Price; twenty on any day it closes below. The Knock-Out is tested on the close, from the trade date onwards.

WORKED EXAMPLE

Same contract, three markets, three very different outcomes

Rally

Knocks out on day 15 — but two periods are guaranteed

Days accumulated 42

Units bought 420 @ 288

Cost USD 120,960

Value at 336 USD 141,120

P&L **+USD 20,160**

Return on margin **+4.7%**

Sideways

Drifts to 300, never knocks out

Days accumulated 250

Units bought 2,500 @ 288

Cost USD 720,000

Value at 300 USD 750,000

P&L **+USD 30,000**

Return on margin **+6.9%**

Fall

Gaps below the Forward Price in week one, slides to 160

Days accumulated 250 (all geared)

Units bought 5,000 @ 288

Cost USD 1,440,000

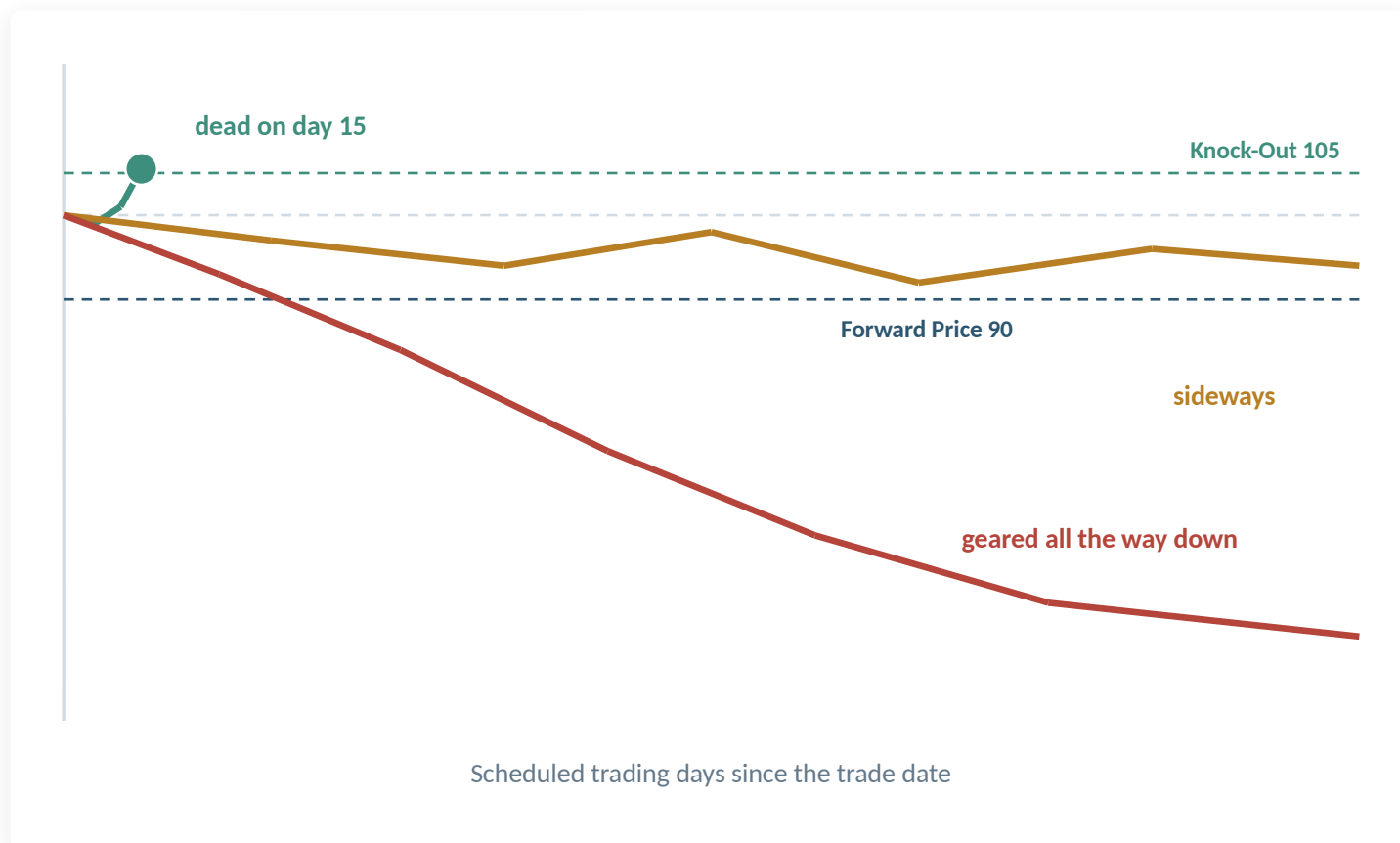
Value at 160 USD 800,000

P&L **-USD 640,000**

Return on margin **-148%**

Best realistic case +6.9%. Bad case -148%. If the share went to zero the client would owe the full USD 1.44m Max. Exposure against roughly USD 432,000 posted — 3.3× the margin.

The knock-out is a trapdoor, not a target



- **The client's good scenarios are short and small.**
- A rally does not make the client rich — it terminates the trade after three weeks with a few percent of margin earned, and the guaranteed periods still run to the end of month two.
- The client's bad scenario is long and large. Nothing stops the accumulation. Every day the stock is lower, the client buys twice as much of it.
- **There is no symmetry here. The trade is designed so that the pleasant outcome ends quickly and the unpleasant one runs to term.**

Decumulators: the mirror image, with no ceiling

Client P&L for a single observation day



The pitch

'Sell your concentrated holding at a premium to today's price.' Attractive to founders and families sitting on one large position.

The catch

If the stock rallies, the client is obliged to sell twice as much, at a price that gets worse every day. A share price has no ceiling.

Why regulators single it out

Accumulator losses are capped at the full notional. Decumulator losses are not capped at all. The SFC and HKMA say so explicitly.

Terms mirror the accumulator: Forward Price 114% of spot, Knock-Out 95%. The client sells 10 units a day below the Forward Price and 20 above it.

A strip of barrier options, priced as one trade

1

Long 1 up-and-out call per day

Struck at the Forward Price, knocked out at 105%. This is the client's small daily gain when the share behaves.

2

Short 2 up-and-out puts per day

Same strike, same knock-out. The barrier sits above spot, so both legs die on a rally. This is where the discount — and the loss — comes from.

3

× 250 scheduled trading days

Each day is its own pair of options with its own expiry. The 'trade' is a portfolio of 750 options.

What that means for risk

Short volatility

Net, the client has sold twice as many options as bought. A vol spike hurts the mark before the stock has moved at all.

Short skew

The puts sold are below spot, where the market charges the most. Skew steepening in a selloff hits the mark again.

Path dependent

Two markets ending at the same price can produce completely different P&L, depending on whether the barrier was touched.

Non-linear in size

Max. Exposure is realised exactly in the loss case. Size grows as the trade goes wrong.

THE QUESTION EVERY CLIENT ASKS

Where does the discount actually come from?

“The share purchase discount price in the stock accumulator contract in fact comes from the premium the investor receives from selling the options to the counterparty.”

Meena Datwani, then Executive Director, Hong Kong Monetary Authority

It is not a discount

It is a payment for risk the client has agreed to take. Priced fairly, its expected value is roughly zero before costs.

It is not dollar-cost averaging

DCA buys a constant amount regardless of price. An accumulator buys double when the price falls and stops entirely when it rises. It is the opposite of DCA.

It is not 'I wanted the stock anyway'

Wanting HK\$3m of a stock is not the same as an HK\$21m obligation to keep buying it. Size, not preference, is what breaks clients.

These three sentences are the entire content of the regulators' concern about how this product is sold.

Most clients are not stopped out by the payoff — they are stopped out by margin

- **The mark-to-market is an option valuation, not intrinsic value.**
- The position can lose value on a rise in implied volatility alone, with the share price unchanged.
- Initial margin is typically 30% of maximum notional for equity accumulators, 10% for FX. Banks reserve the right to raise it — up to 100% — during the life of the trade.
- If the client cannot meet the call, the bank closes out at its own price. There is no exchange, no screen, and no obligation to unwind gradually.
- **This is how most of the 2008 losses were actually realised: forced unwind cost, not accumulated shares.**

Shine Grace Investment Ltd v Citibank

Nine equity accumulators bought over two days in October 2007. Three knocked out within a month. The remaining six were unwound in January 2008.

HK\$427m

Unwind cost

Paid to close the six surviving contracts early

HK\$51m

Loss on shares

The part that came from actually holding the stock

The pitch and the payoff, side by side

What the client hears

"Buy the stock you like at a 12% discount."

"It's disciplined phased entry."

"Capital efficient — only 30% down."

"You have a range-bound view."

What they have actually agreed to

You are paid a premium to accept a leveraged obligation to keep buying it while it falls.

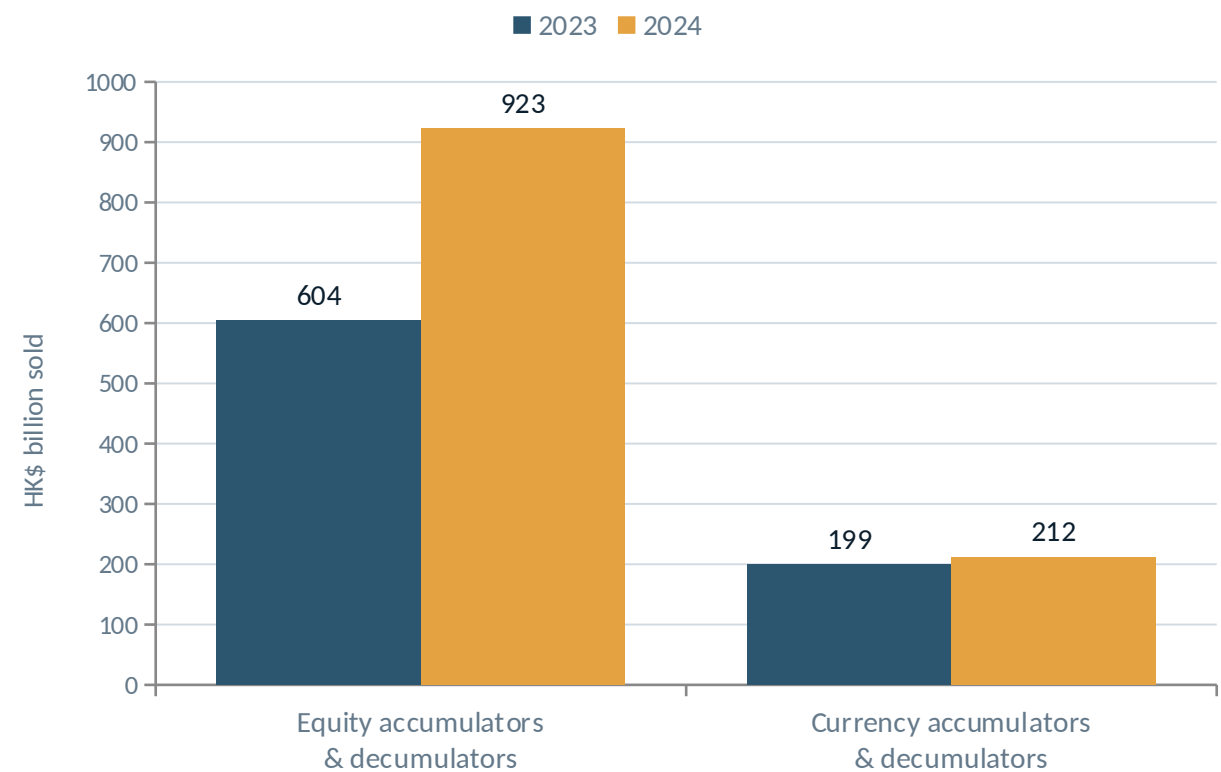
Phased entry stops when you want it to. This stops only when you are winning.

Leverage of 3.3×. The margin is the smaller number on the page, not the risk.

Correct — and this is one of the only instruments that monetises that view. It is not a bad product, it is a misunderstood one.

The last row is the important one. Accumulators are a legitimate way to express a sideways view. Almost nobody who loses money on one had that view.

This is not a niche product in Asia



44%

of HK structured products

Accumulators and decumulators combined, 2024

1.2m

clients, all product types

Buyers of non-exchange-traded products, all types; 90% not PIs

≈ HK\$1.1 trillion

of accumulators and decumulators sold in Hong Kong in 2024 alone — against roughly US\$23bn outstanding at the 2008 peak.

Source: SFC–HKMA Joint Product Survey 2023 and 2024. Figures are annual sales notional, not outstanding positions.

IT HAS GONE WRONG BEFORE

Three episodes worth knowing by name

2008

Hong Kong

Roughly US\$23bn of accumulators outstanding by April 2008; far more once gearing is counted. Individual private-bank losses ran into the hundreds of millions of HK dollars. The product acquired its nickname here.

2014–17

Taiwan TRFs

Leveraged CNH target-redemption forwards — FX accumulators in all but name, sold to exporters as hedging. After the August 2015 renminbi devaluation, Taiwanese SMEs were estimated to have lost around US\$6.6bn.

Jan 2024

China snowballs

A different product with the same DNA. Around US\$50bn notional outstanding with roughly 40% of knock-in levels already breached; hedge unwinds pushed CSI 1000 futures limit-down on 22 January.

The common thread is not the asset class. It is leverage, a capped upside, and a client base that read 'discount' as 'safe'.

Clients lose in court. Banks lose to regulators.

- **Hong Kong courts have consistently found for the banks.**
- Shine Grace v Citibank (HK Court of Appeal, 2022): a HK\$478m accumulator loss, claim dismissed. Volunteering advice did not create a duty to advise on suitability where the contract said otherwise.
- Deutsche Bank v Chang Tse Wen (Singapore Court of Appeal, 2013): a US\$49m first-instance award to the client, reversed in full on appeal.
- **The exposure is regulatory and remediation-driven, not litigation-driven.**
- HSBC Private Bank (Suisse) was fined HK\$400m in 2017 — then a Hong Kong record — over 2003–08 sales that included forward accumulators.

What changed recently

HKMA, 14 March 2025

A single consolidated Guidance on Selling of Accumulators replaced seven legacy circulars. No new obligations, but it is now the one document to read.

SFC / HKMA, July 2023

The Sophisticated Professional Investor regime — HK\$40m portfolio or HK\$80m net assets plus a knowledge test — streamlines the transaction-level suitability requirements. Much accumulator flow now sits under it.

Joint thematic review, 2024

Found firms failing to capture leverage and volatility in product due diligence, and recorded a salesperson unable to explain accumulator mechanics beyond reading the term sheet aloud.

Five things about accumulators

- 1 It is an obligation to buy, not an option to buy. Every trading day, whatever happens.
- 2 The discount is option premium the client has sold. It is not a bargain and it is not carry.
- 3 Gearing means the client buys most of the position in exactly the scenario they did not want.
- 4 Maximum notional, not margin, is the size of the trade. Typically 3–4× the cash posted.
- 5 The upside ends early and small; the downside runs to term and large. That asymmetry is the entire product.